INSTITUTE OF SONOLOGY

Royal Conservatoire The Hague

Chaos

Composing with Algorithms
http://www.bjarni-gunnarsson.net

Dynamical Systems

A **dynamical system** can be said to be anything that has **motion**. For example a swinging pendulum, a bouncing ball or flowing water.

The **motion** of a dynamic system is usually dependent on how it <u>changes in time</u>.

Deterministic dynamic systems are governed by a <u>fixed set of rules</u> that describe how it evolves from one state to the next.

These **rules** can usually be represented in a <u>model by equations</u> that are parameterized by time and the previous state of a system.

Dynamical Systems

"DYNAMICAL SYSTEMS THEORY (or dynamics) concerns the description and prediction of systems that exhibit complex changing behavior at the macroscopic level, emerging from the collective actions of many interacting components. The word dynamic means changing, and dynamical systems are systems that change over time in some way.

Dynamical systems theory describes in general terms the ways in which systems can change, what types of macroscopic behavior are possible, and what kinds of predictions about that behavior can be made."

(Melanie Mitchell)

Prediction

The general study of motion is called **mechanics**.

Isaac Newton pioneered work in classical mechanics. Mechanics is divided into two areas: **kinematics**, which describes how things move, and **dynamics**, which explains why things obey the laws of kinematics.

Newtonian mechanics produced a picture of a *"clockwork universe"*. The mathematician Pierre Simon Laplace asserted that, given Newton's laws and the current position and velocity of every particle in the universe, it was possible, in principle, <u>to predict everything for all time.</u>

Chaos

The discovery of **chaos** led to the knowledge that perfect prediction of all complex systems is impossible.

Chaotic systems are the ones where even minuscule uncertainties in measurements of initial position and momentum can result in huge errors in long-term predictions of of these quantities. This is known as *"sensitive dependence on initial conditions."*

Even the tiniest error in an initial measurement will eventually produce huge errors in your prediction of the future motion of an object.

Prediction

"If we knew exactly the laws of nature and the situation of the universe at the initial moment, we could predict exactly the situation of that same universe at a succeeding moment. But even if it were the case that the natural laws had no longer any secret for us, we could still only know the initial situation approximately. If that enabled us to predict the succeeding situation with the same approximation, that is all we require, and we should say that the phenomenon has been predicted, that it is governed by laws. But it is not always so; it may happen that small differences in the initial conditions produce very great ones in the final phenomenon. A small error in the former will produce an enormous error in the latter. Prediction becomes impossible…."

Henri Poincaré

Motion Types

The simplest type of motion is a **fixed point behavior** (for example a halted pendulum or ball that has stopped.)

The second type of behavior is **periodic motion** where a certain movement is repeated over and over.

Quasi-periodic systems contain a slightly more complex behavior where they behave similarly to periodic systems but never quite repeat themselves.

The fourth type of deterministic dynamical systems are **chaotic systems**.

Iterative Processes

An **iterative process** is the repeated application of a mathematical equation where at each step previous values are used to produce the ones for the next step.

Iterative processes are used to describe how dynamical systems in nature move and change with time.

An iterative process can under certain conditions produce **chaotic behavior**.

Most of the mathematics of chaos involve repeated processes of simple mathematic formulas.

Orbits & Phase Spaces

The traced output of an iterative process is referred to as its **orbit**.

An **orbit** is created for a set of input values and varies with different inputs.

Orbits fall into the three categories: **fixed**, **periodic** and **chaotic**.

The **phase space** is defined as the area in which the dynamics of a system happen.

Knowing the phase space means knowing the space where all possible configurations of a system will occur.

The Logistic Map

A **linear system** is one you can understand by understanding its parts individually and then putting them together. A **nonlinear system** is one in which the whole is different from the sum of the parts.

The **logistic map** is a simple population growth model. It is set to capture reproduction of a small population or starvation of a bigger one.

The **logistic model** is simplified by combining the effects of birth rate and death rate into one number, called R. Population size is replaced by a related concept called *“fraction of carrying capacity”* called x.

The Logistic Map

The **logistic map** is defined by an iterative equation:

$\mathbf{x_{n+1} = r * x_n * (1 - x_n)}$

(where r is a parameter that reflects the reproduction rate).

Depending on the value of r, the logistic map can reach **stable**, **periodic** and **chaotic** states.

The Logistic Map

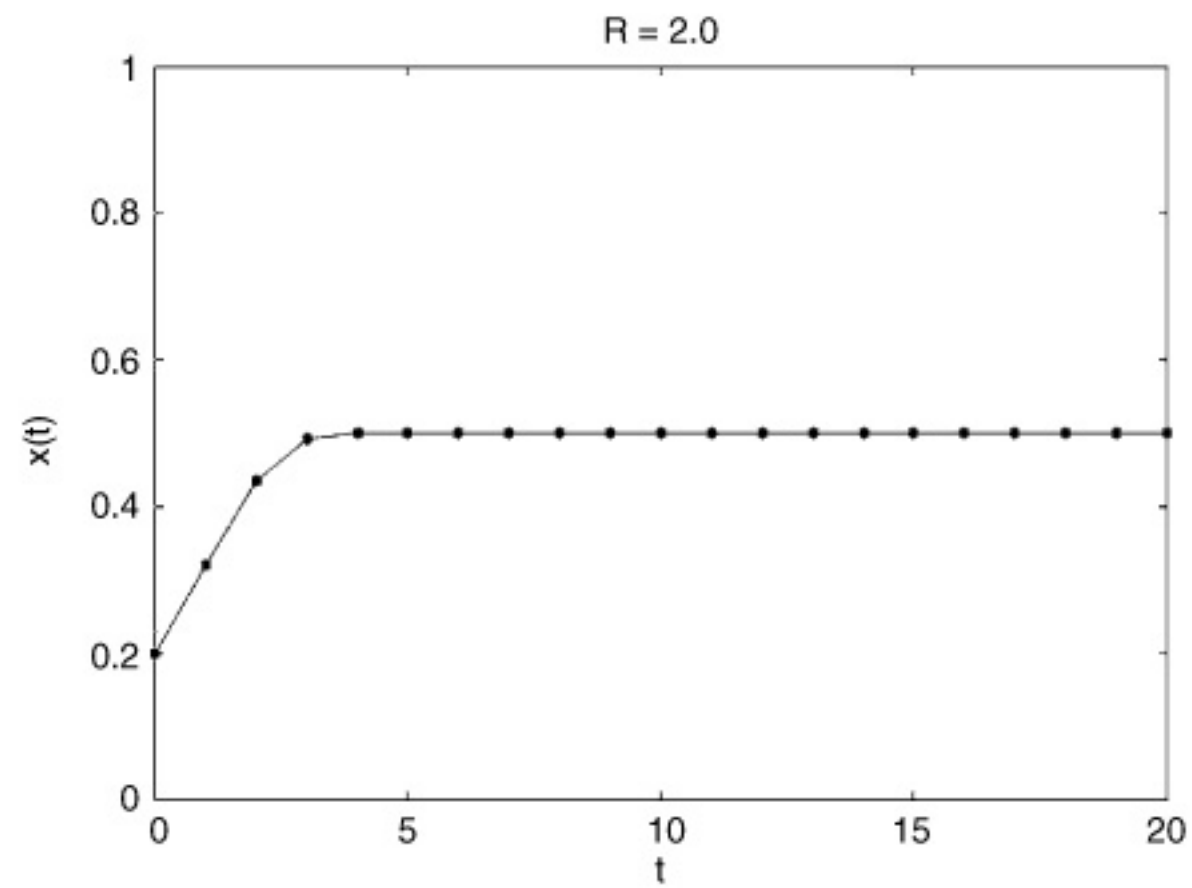


FIGURE 2.6. Behavior of the logistic map for $R = 2$ and $x0 = 0.2$.

(From ‘Complexity’ by Mitchell)

The Logistic Map

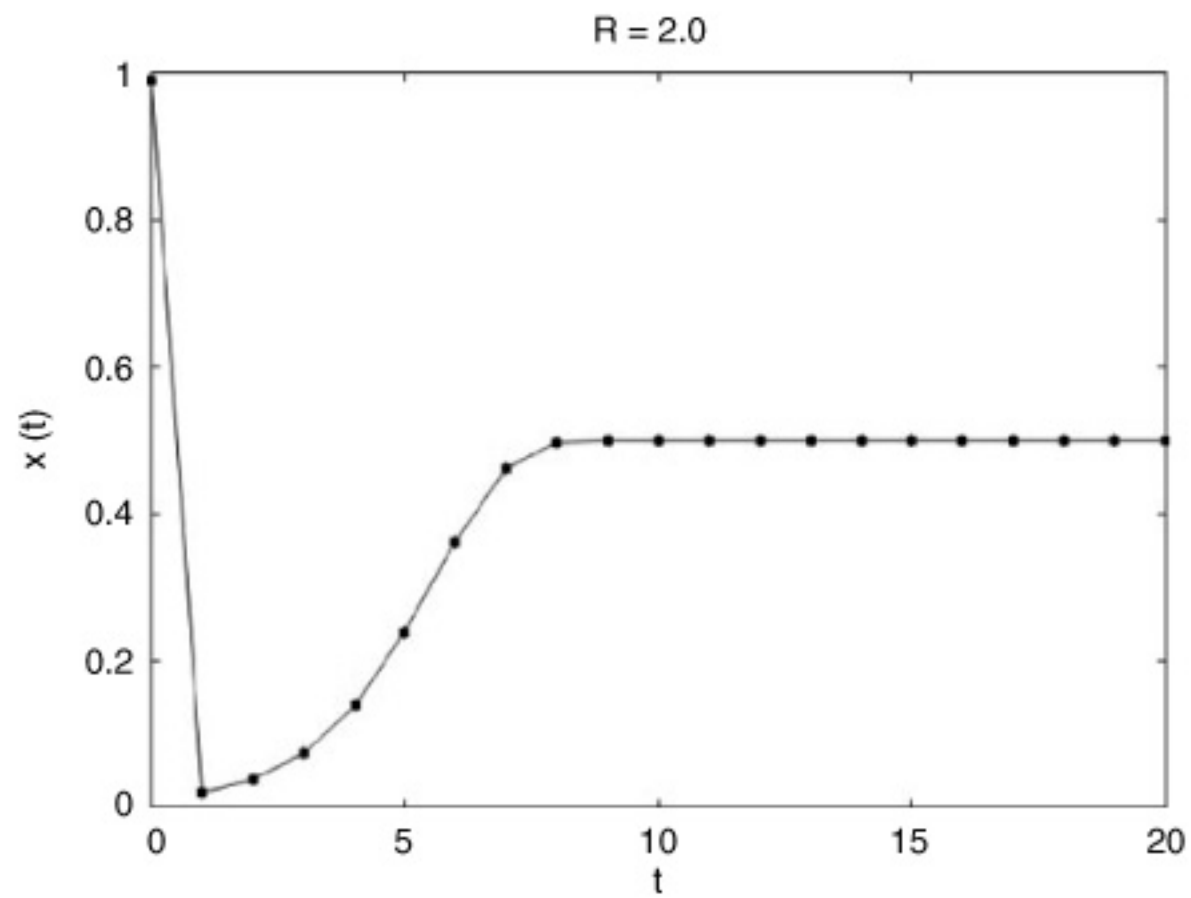


FIGURE 2.7. Behavior of the logistic map for $R = 2$ and $x0 = 0.99$.

(From 'Complexity' by Mitchell)

The Logistic Map

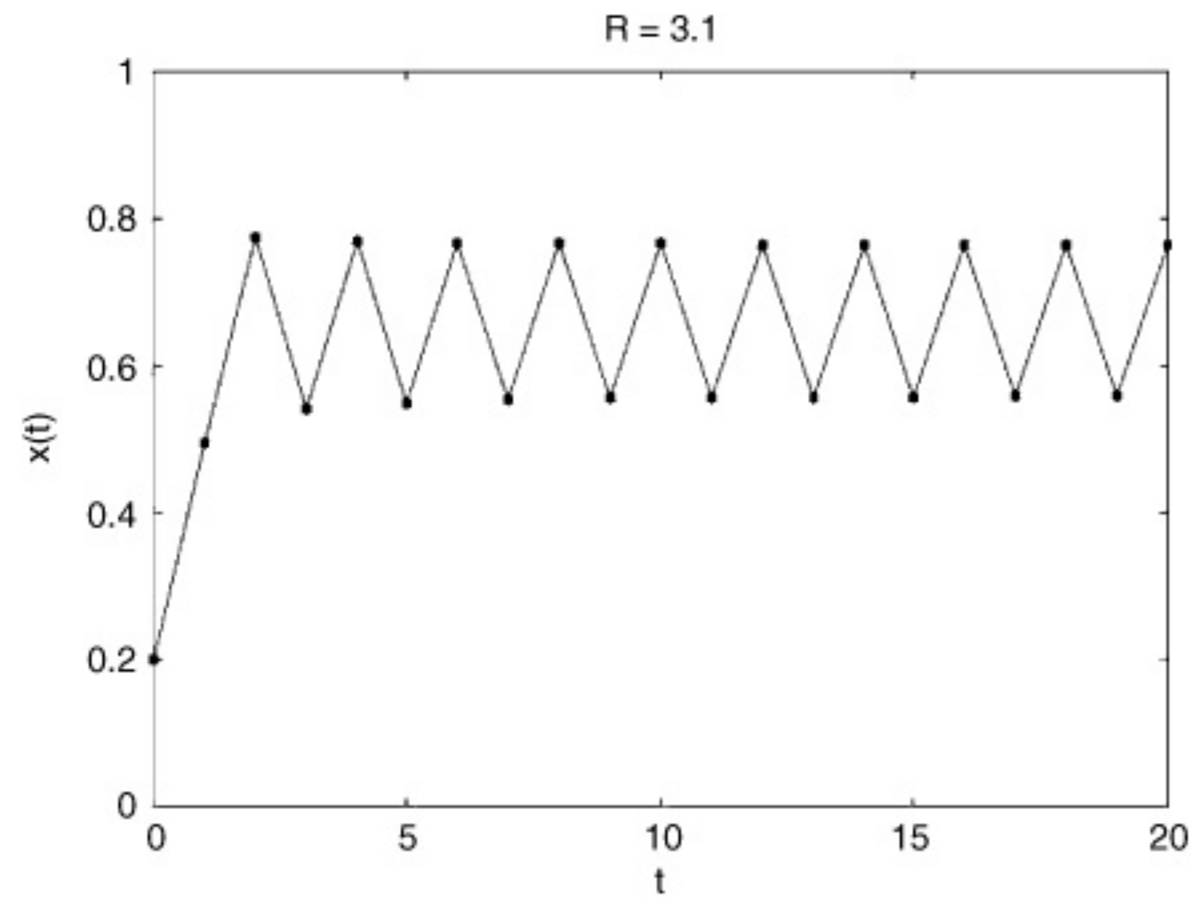


FIGURE 2.8. Behavior of the logistic map for $R = 3.1$ and $x0 = 0.2$.

(From 'Complexity' by Mitchell)

The Logistic Map

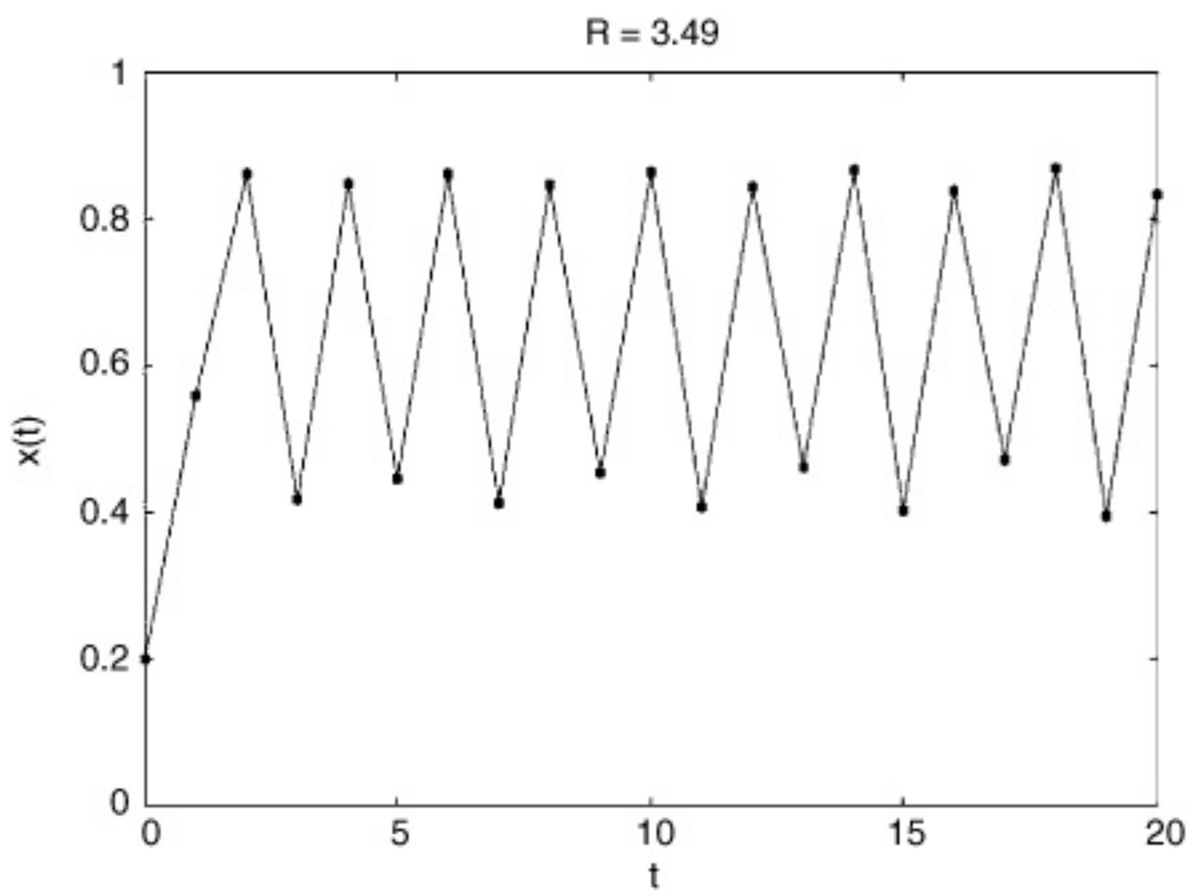


FIGURE 2.9. Behavior of the logistic map for R = 3.49 and x0 = 0.2.

(From 'Complexity' by Mitchell)

The Logistic Map

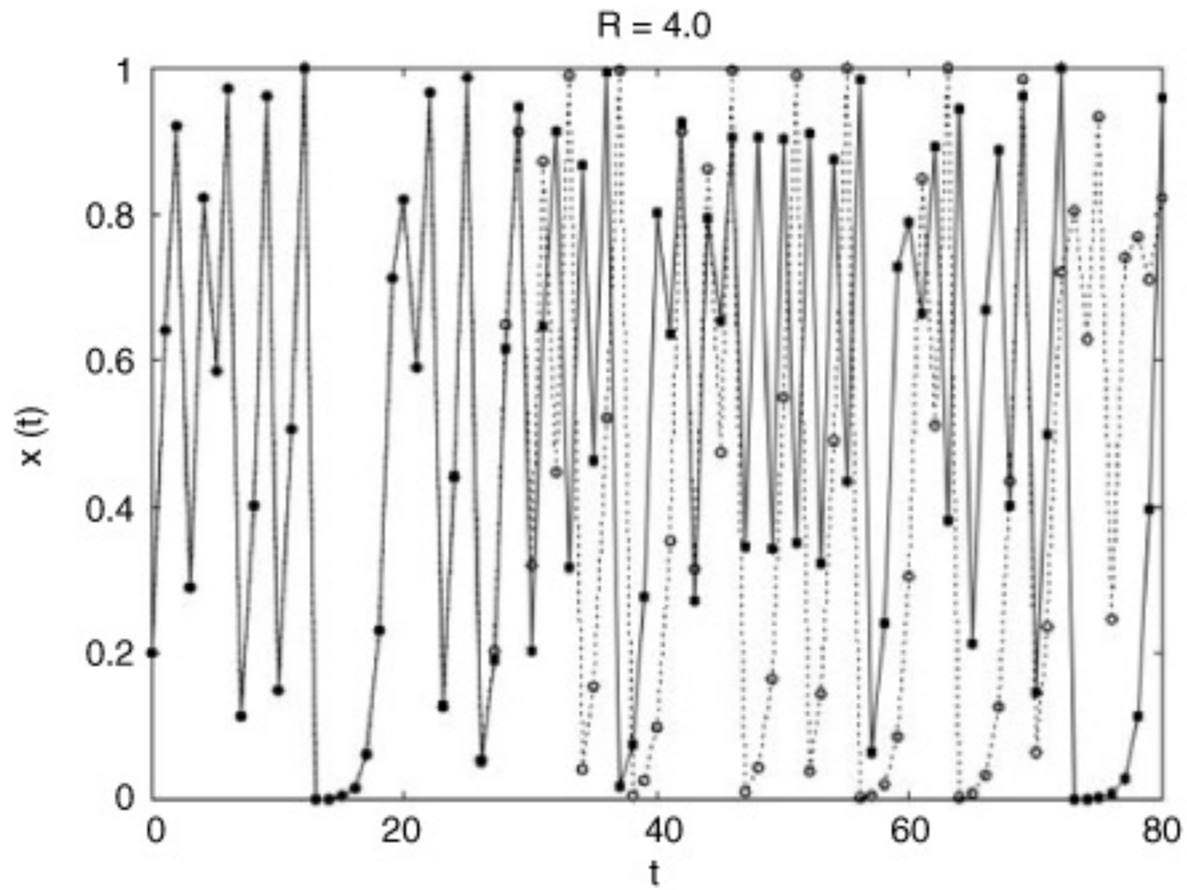
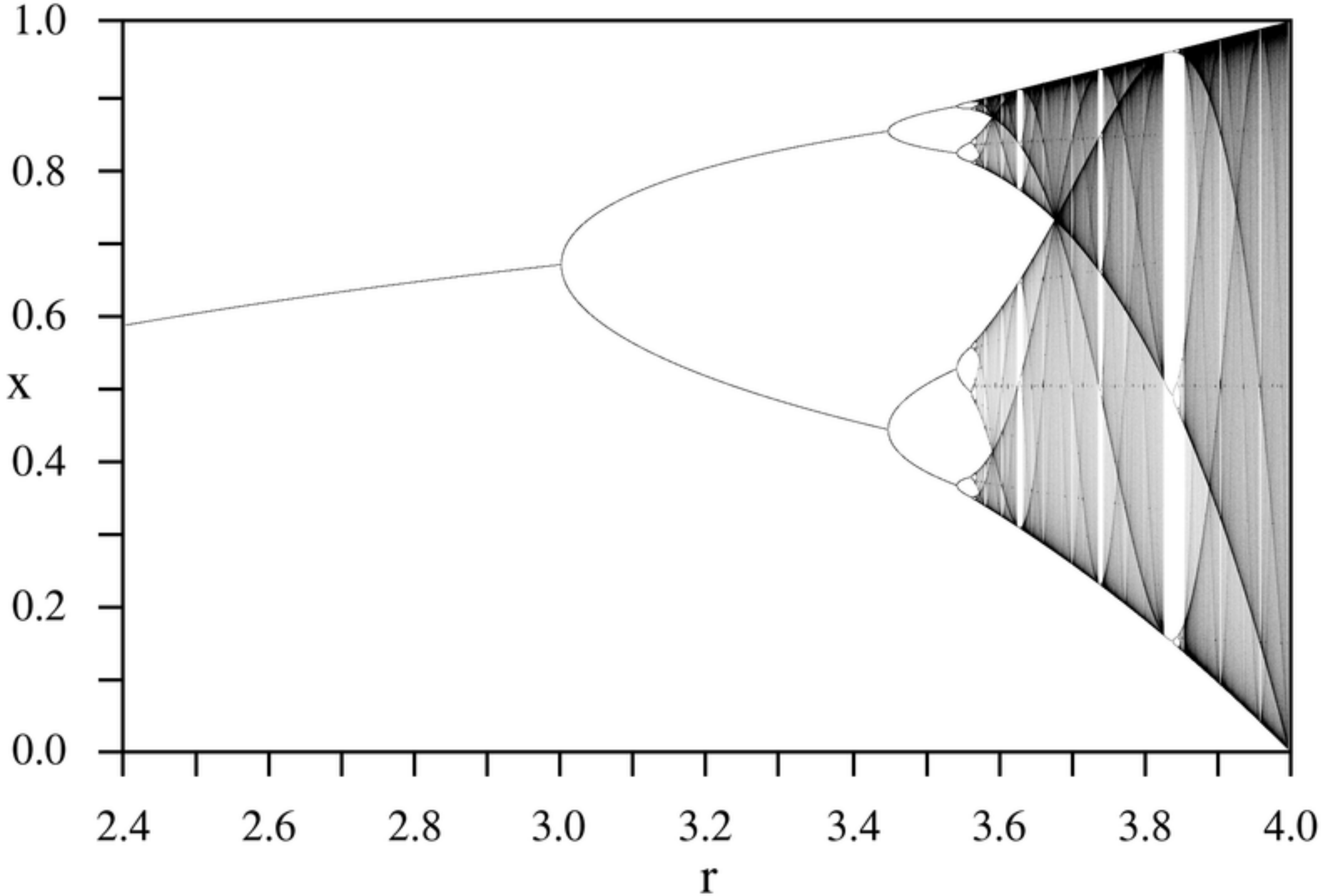


FIGURE 2.10. Two trajectories of the logistic map for $R = 4.0$: $x0 = 0.2$ and $x0 = 0.2000000001$.

(From 'Complexity' by Mitchell)

The Logistic Map



(A bifucuration diagram of the logistic map.)

Chaos

The word chaos is usually used to discuss a sense of "*disorder*". In mathematics it is used to describe **deterministic** systems that are **hard to predict** and **sensitive to initial conditions**.

The most important difference between **chaotic** and **stochastic** processes is that the future behavior of a chaotic system can be <u>predicted</u> in the short term while stochastic process can only be statistically characterized.

Chaos

Due to **sensitive dependance** on **initial conditions**, chaotic systems can be impossible to predict in the long term.

There is still some *"order in chaos"* reflected in common properties of these systems making some high-level aspects (period-doubling) possible to predict.

Bifurcation

A **bifurcation** is a qualitative change in the behavior of a dynamic system as a parameter is varied.

Essentially, it's a point where the system **transitions** from one type of behavior to another.

Bifurcations are often associated with the **emergence** of complex and unpredictable dynamics, including chaotic behavior.

Characteristics of Chaos

*** Deterministic.**

Chaotic processes are not random. There always exists a structure in their phase space.

*** Sensitive.**

A chaotic system is extremely sensitive to initial conditions. This means a slight variation of initial input can have a strong effect of its long term behavior.

*** Ergodic.**

A state space trajectory will always return to the local region of a previous point in the trajectory. Having the same weather twice.

*** Embedded.**

Chaotic attractors are embedded with an infinite number of unstable periodic orbits.

Paths to Chaos

*** Period Doubling**

A fixed point loses stability and a sudden doubling of the repetition period of the original system occurs.

*** Intermittency**

The alternation of phases of regular and chaotic dynamics. A process gets disturbed again and again until chaos takes over.

*** Turbulence**

A flow process determined by chaotic property changes. Turbulence is a dissipative process that is always chaotic. Not all chaotic processes are turbulent.

Dissipative and Conservative Systems

*** Dissipative Systems**

Where the phase space of an orbit shrinks in time. This shrinkage leads to the formation of an attractor as time unfolds and after an initial period on transients.

*** Conservative Systems**

Where the phase space remains constant and orbits are not drawn to an attractor or traverse a transient phase.

Strange Attractor

Long-term behavior of dynamical systems often approaches a set of possible solutions.

This means that the phase space gets more narrow as time unfolds. The phase space is then determined by its **attractor**.

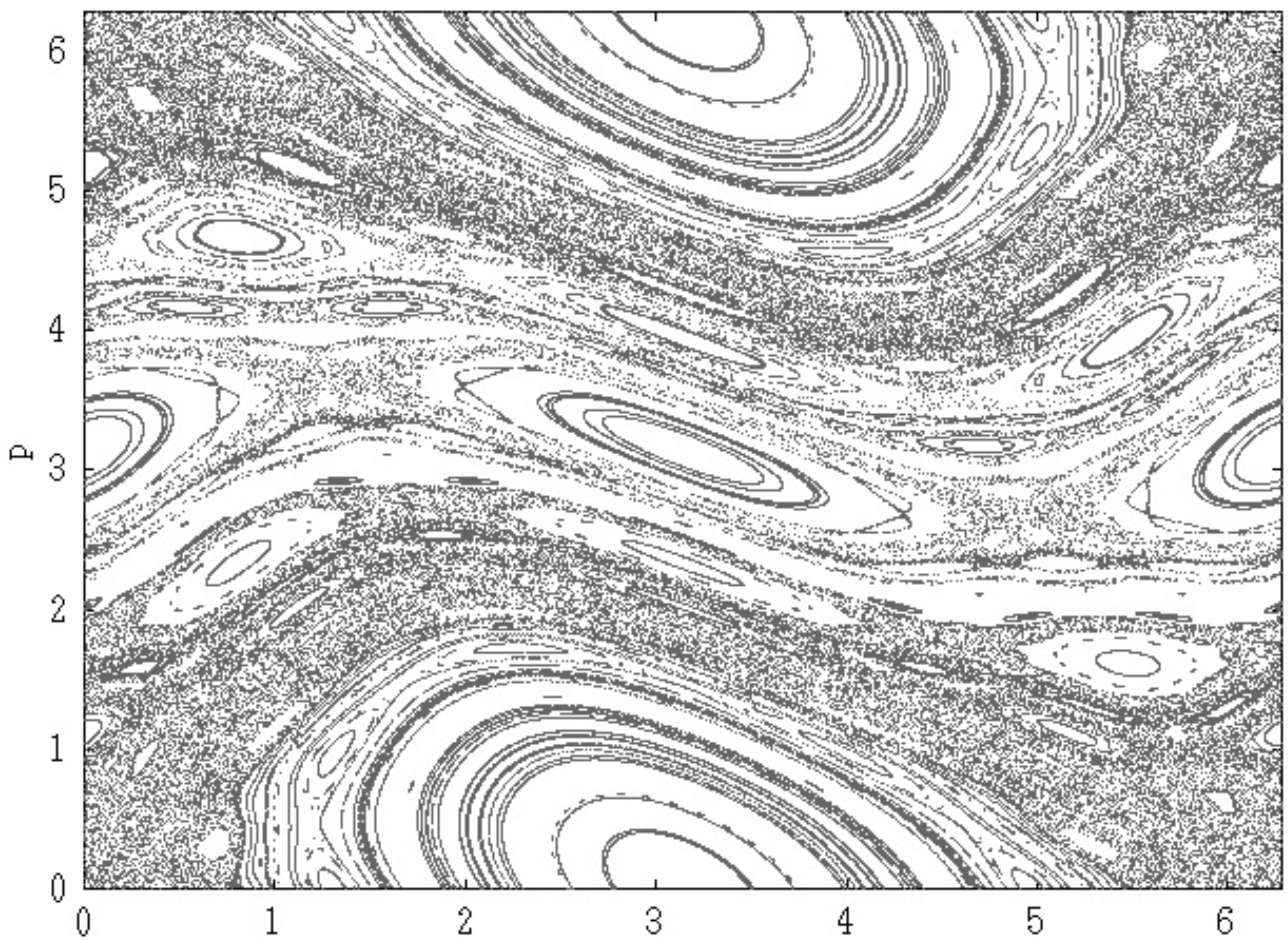
Simple attractors exist such as a fixed point for a pendulum. Another possibility is limit cycle, a sequence of values repeated periodically.

For many chaotic systems more **complex attractors** exist. These do not have an simple integer dimension and therefore exist as fractals. These are called **strange attractors**.

The Standard Map

An example of a conservative and iterated map.

Derived from a model of a pendulum that receives external periodic perturbations.



Lorenz System (or butterfly attractor)

Is a **dissipative** system in <u>three dimensions</u> with chaotic solutions for certain input values.

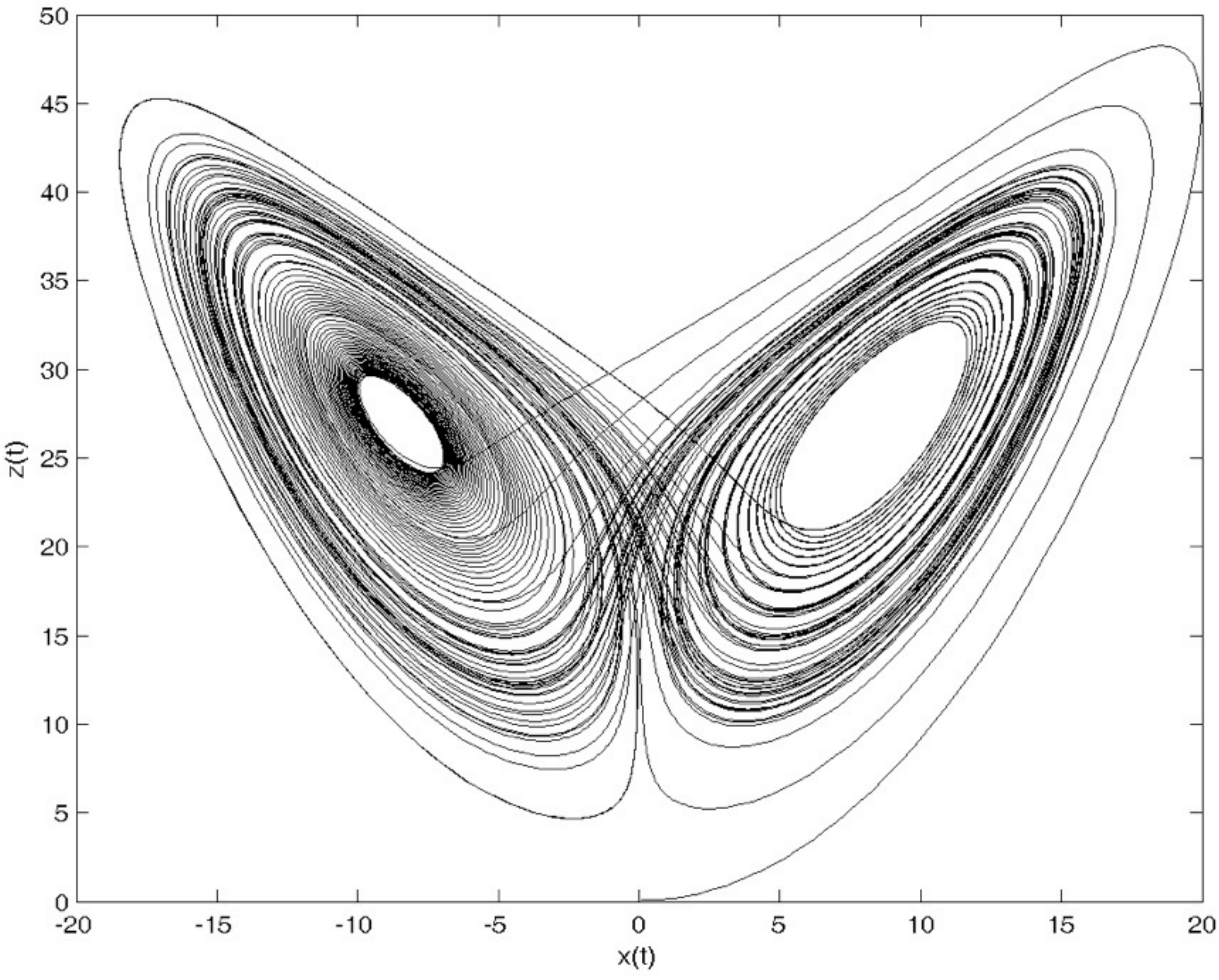
Was derived from a simplified model of convection in the earths atmosphere.

Displays continuous flows where orbits are smooth, unbroken curves.

The **attractor**, when plotted, looks like a butterfly or the figure eight.

$$dx / dt = a (y - x)$$
$$dy / dt = x (b - z) - y$$
$$dz / dt = xy - c z$$

The Lorenz Attractor



Ideas from Chaos

“Seemingly random behavior can emerge from deterministic systems, with no external source of randomness.

The behavior of some simple, deterministic systems can be impossible, even in principle, to predict in the long term, due to sensitive dependence on initial conditions.

Although the detailed behavior of a chaotic system cannot be predicted, there is some “order in chaos” seen in universal properties common to large sets of chaotic systems, such as the period-doubling route to chaos and Feigenbaum’s constant. Thus even though “prediction becomes impossible” at the detailed level, there are some higher-level aspects of chaotic systems that are indeed predictable.”

(Melanie Mitchell)

Musical Properties

Structures that benefit of mapping from of chaos theory are in most cases represented musically by means **transferring trajectories** of a phase space on musical parameters.

Once the underlying equation system has been chosen and the initial values have been set, the system continuously produces new events whose progression is hard to predict.

In order to receive different but nevertheless correlated values for musical parameters, complex mapping strategies for equation systems are needed.

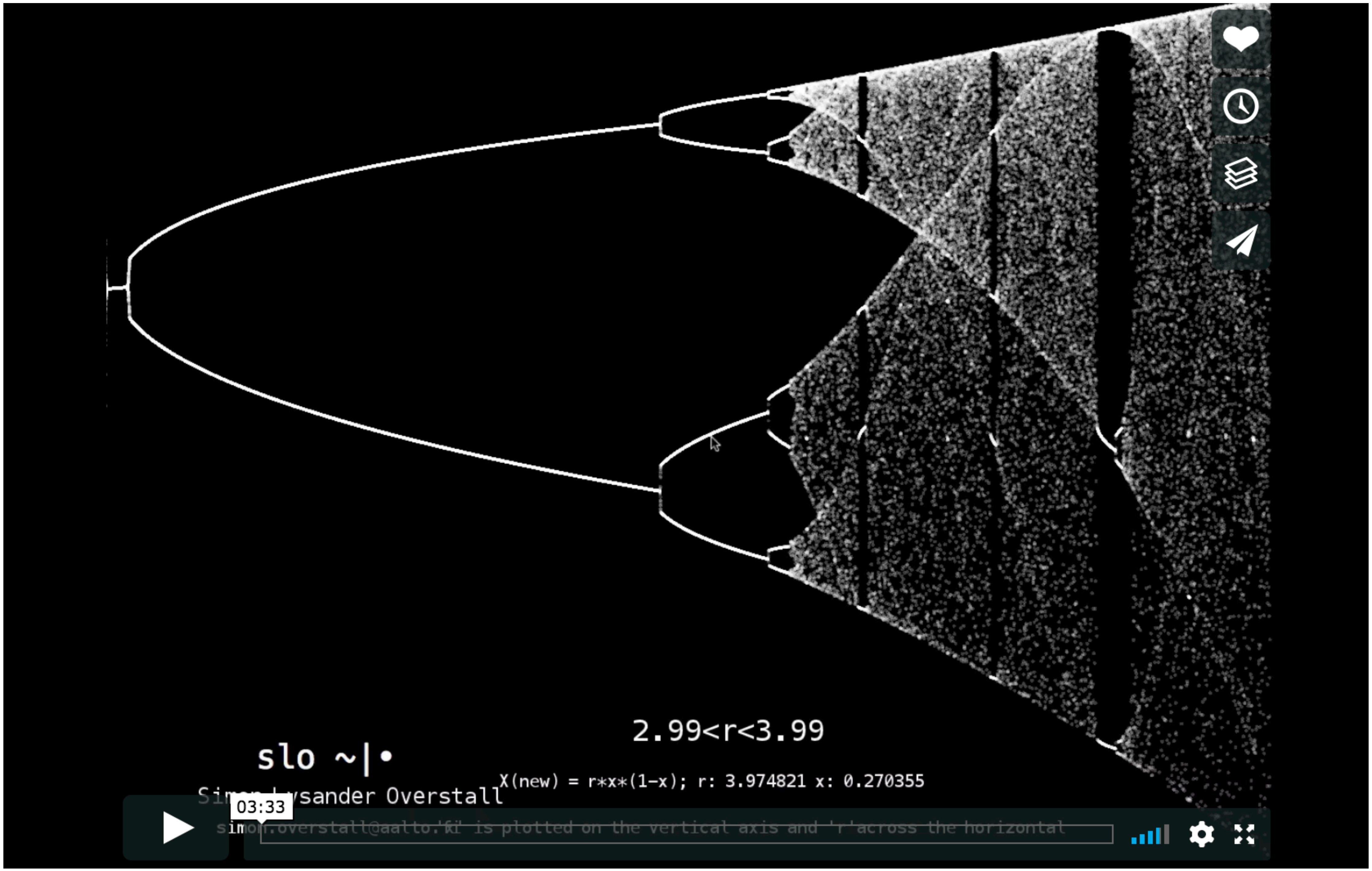
Musical Properties

Chaos contains an interesting mix of **order** (or periodicity) and **disorder**.

Strange attractor can function as a set of possibilities where you have something that is determined but unpredictable.

The item represented on the attractor are potentials in a musical context other events can not occur. This is different from probabilities where everything can happen.

Order in music becomes synonymous with the effect of perceptual continuity. The boundaries between **order** and **disorder** can be interesting to explore and chaos theory provides powerful tools to do so.


2.99<r<3.99
slo ~|•
X(new) = r*x*(1-x); r: 3.974821 x: 0.270355
ysander Overstall
03:33
'x' is plotted on the vertical axis and 'r' across the horizontal

Practice

Nonlinear Dynamics In Musical Interactions

"Many composers and musicians have employed nonlinear dynamical systems explicitly in their work, implementing a variety of systems for a variety of purposes." […] "Several features of such systems are highlighted as being of potential musical interest: fixed points, limit cycles, bifurcations, chaos and strange attractors." […] " The outputs from the Logistic map and several other nonlinear dynamical systems were mapped directly to particular compositional parameters such as frequency, volume, timing, attack, and so on, and iterated at various tempos to create compositions. Pressing is particularly interested in regions that are only intermittently chaotic, that oscillate between quasi-periodic stability and chaotic states, and thus chooses values of r accordingly"

Tom Mudd

Complex Musical Behaviours

"Chaotic systems can offer possibilities for music composition both at timbre level or formal level. Di Scipio has explored these techniques extensively for the generation of timbres." […] "The behaviour of chaotic systems depends on internal parameters that regulate the nonlinearity, as well as on the initial conditions that trigger the systems. It is possible to identify at least four fundamental behaviours in chaotic systems: fixed-point attractor, periodic oscillation, strange attractor, chaos"

Dario Panfilippo

Agostino Di Scipio & Dario Sanfilippo - Contexture 1 (2019)

Exercises

```
(
	NF(\iop, {|freq=78, mul=1.0, add=0.0|
		var noise = LFNoise1.ar(0.001).range(freq, freq + (freq * 0.1));
		var osc = SinOsc.ar([noise, noise * 1.04, noise * 1.02, noise * 1.08],0,0.2);
		var out = DFM1.ar(osc,freq*4,SinOsc.kr(0.01).range(0.92,1.05),1,0,0.005,0.7);
		HPF.ar(out, 40)
	}).play;
)
```

```
(
	NF(\dsc, {|freq = 1080|
		HPF.ar(
			BBandStop.ar(Saw.ar(LFNoise1.ar([19,12]).range(freq,freq*2), 0.2).excess(
			SinOsc.ar( [freq + 6, freq + 4, freq + 2, freq + 8])),
			LFNoise1.ar([12,14,10]).range(100,900),
			SinOsc.ar(20).range(9,11)
		), 80)
		;
	}).play;
)
```

```
var <>pindex, <>cindex;

initialize {
	if(pindex.isNil, { pindex = 1000 });
	if(cindex.isNil, { cindex = 2000 });
}

clearProcessSlots {
	pindex = 1000;
	(this.pindex - 1000).do{|i| this[this.pindex+i] = nil; }
}

clearOrInit {|clear=true|
	if(clear == true, { this.clearProcessSlots() }, { this.initialize() });
}

transform {|process, index|
	if(index.isNil && pindex.isNil, {
		this.initialize();
	});

	pindex = pindex + 1;
	this[pindex] = \filter -> process;
}

control {|process, index|
	var i = index;

	if(i.isNil, {
		this.initialize();
		cindex = cindex + 1;
		i = cindex;
	});

	this[i] = \pset -> process;
}
```

```
(
	NF(\depfm, {|freqMin=5, freqMax=20, mul=20, add=80, rate=0.5, modFreq=2100, index=0.3, amp=0.2|
		var trig, seq, freq;
		trig = Dust.kr(rate);
		seq = Diwhite(freqMin, freqMax, inf).midicps;
		freq = Demand.kr(trig, 0, seq);
		HPF.ar(PMOsc.ar(LFCub.kr([freq, freq/2, freq/3, freq/4], 0, mul, add),
		LFNoise1.ar(0.3).range(modFreq,modFreq*2), index) * amp, 50)
	}).play;
)
```

Exercises

1. Implement parallel chaotic streams that investigate the deviations of an equation in time. Each stream should have a clear path of observation for example cover a certain frequency region.

2. Explore the difference between using chaotic functions for sound and event control. For example chaotic patterns could be used for rhythmic events while chaotic UGens could control sound properties such as frequency and amplitude.

3. Triggers in SuperCollider can be used for various control. Implement a process that uses chaos-based triggers to create very fast event generation approaching the microtime of granular synthesis.

4. Extend the iterative functions included for sound control.